1. Introduction

This project explores the effectiveness of transformer-based architectures for multi-class text classification on the 20 Newsgroups dataset. Pre-trained models, including BERT, RoBERTa, and DistilBERT, are fine-tuned and systematically compared under different experimental settings.

Beyond model comparison, this study investigates how traditional preprocessing techniques and data augmentation strategies influence performance in a transformer context. Given that transformers are pretrained on large-scale raw text corpora, it is not immediately clear whether conventional NLP practices such as stopword removal or stemming remain beneficial. This project aims to provide empirical evidence to address this question.

Additionally, interpretability is examined using attention-based analysis and model-agnostic methods, and the final model is exported in ONNX format to demonstrate deployment readiness.

2. Dataset and Preprocessing

The 20 Newsgroups dataset consists of approximately 18,000 documents spanning 20 distinct categories, including politics, religion, sports, and computer-related topics. The dataset was split into training, validation, and test sets using stratified sampling to preserve class distributions.

In the baseline setting, minimal preprocessing was applied, and the raw text was preserved. This design choice reflects how transformer models are pretrained, allowing them to fully leverage contextual information.

To further explore preprocessing effects, an alternative pipeline was introduced, incorporating stopword removal and stemming. While these techniques are commonly used in traditional NLP workflows, their compatibility with transformer models remains an open question, which is addressed in this study.

3. Model Architecture and Training

Three transformer models were evaluated: BERT, RoBERTa, and DistilBERT. Each model was fine-tuned with different classification heads, including a linear classifier, a multi-layer perceptron, and an attention-based pooling layer.

Training was performed using the AdamW optimizer with a learning rate scheduler. Mixed-precision training was employed to improve computational efficiency, particularly when running on GPU. Early stopping based on validation performance was used to prevent overfitting, and multiple checkpoints were saved throughout training. The training setup was designed to balance performance and computational efficiency while allowing for meaningful comparison across configurations.

4. Evaluation and Results

4.1 Overall Performance

Model performance was evaluated using accuracy, precision, recall, and F1 score. The results for the three transformer models are summarized below:

Model	Accuracy	Macro F1	Weighted F1
BERT	0.754	0.739	0.752
RoBERTa	0.750	0.734	0.738
DistilBERT	0.746	0.729	0.741

While BERT achieves the highest accuracy, it also has the highest macro-F1 score. This suggests that BERT performsdagais wsm better on more frequent categories.

4.2 Confusion Matrix Analysis

To better understand model behavior beyond aggregate metrics, confusion matrices were analyzed under different experimental conditions.

Baseline Model Performance

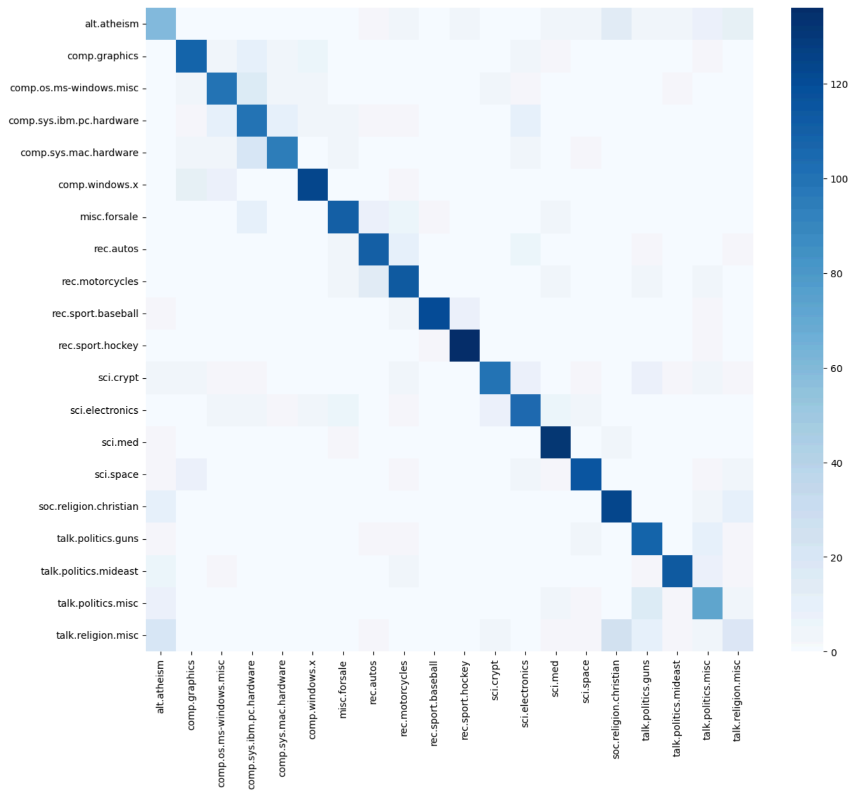

Figure 1: Confusion matrix of the baseline model

As shown in Figure 1, the confusion matrix of the baseline model exhibits a strong diagonal structure, indicating that most samples are correctly classified. This confirms that the transformer model effectively captures the semantic distinctions across categories.

However, a closer inspection reveals that misclassifications tend to occur between semantically similar classes. For example, technical categories related to computer hardware show some overlap, as do politically oriented categories. This pattern suggests that while the model performs well at distinguishing broad topics, it struggles with finer-grained distinctions where vocabulary and context overlap.

Preprocessing Analysis

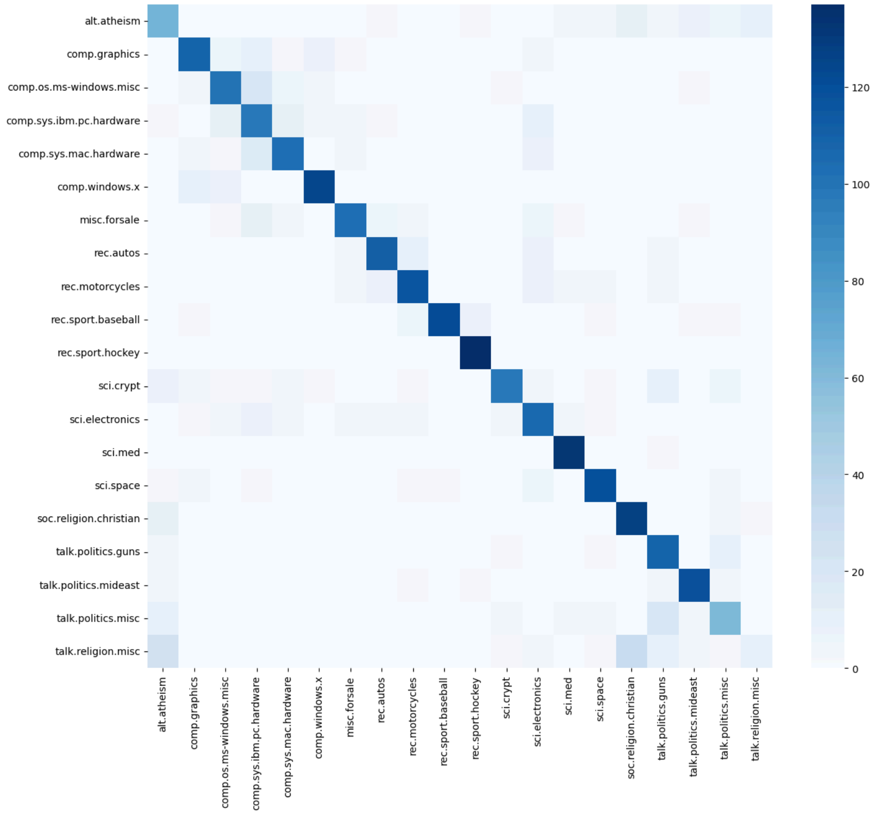

Figure 2: Confusion matrix with stopword removal and stemming

To evaluate the impact of traditional preprocessing, stopword removal and stemming were applied to the input text.

Compared to the baseline (Figure 1), the confusion matrix in Figure 2 shows a slightly weaker diagonal pattern, indicating a reduction in classification accuracy. The macro-F1 score decreased from 0.735 (baseline DistilBERT) to 0.718 after applying stopword removal and stemming. This aligns with the observed drop in macro-F1 score, suggesting that preprocessing negatively affects model performance. An explanation of this situation could be that transformer models rely heavily on contextual relationships between tokens. Removing stopwords may eliminate important syntactic signals, while stemming alters word forms in a way that conflicts with subword tokenization. Since these models are pretrained on natural, unprocessed text, modifying the input distribution can reduce their effectiveness.

In conclusion, this experiment highlights that traditional preprocessing techniques may not only be unnecessary but potentially harmful when applied to transformer-based models.

Data Augmentation Analysis

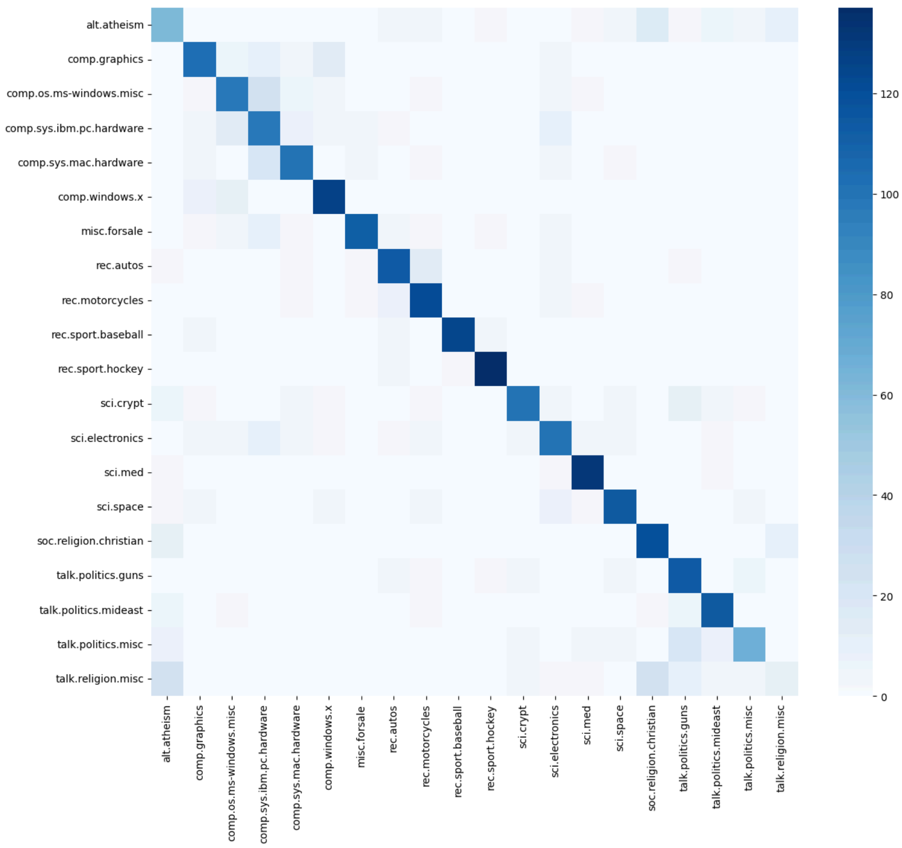

Figure 3: Confusion matrix with data augmentation

To improve generalization, data augmentation was applied using a synonym replacement strategy, increasing the training set from 12,830 to 14,754 samples.

As shown in Figure 3, the confusion matrix exhibits a slightly more concentrated diagonal compared to the baseline. The macro-F1 score improved from 0.735 in the baseline setting to 0.741 after applying data augmentation. This suggests that the model benefits from the increased diversity in training data, leading to improved generalization.

Although the performance gain is modest, it is consistent across metrics. The relatively small improvement can be attributed to the already large dataset size and the strong prior knowledge embedded in pretrained transformer models. Additionally, synonym replacement may introduce noise when substitutions are not perfectly context-aligned.

Nevertheless, the results indicate that even simple augmentation techniques can provide incremental improvements when applied carefully.

5. Interpretability

To better understand model predictions, attention weights were analyzed to identify influential tokens. The model consistently assigns higher attention to semantically meaningful words, particularly domain-specific terms, suggesting that it captures relevant contextual information.

In addition, LIME was incorporated into the pipeline to generate local explanations. However, due to environment constraints, full execution was limited. As an alternative, attention-based analysis was used to interpret model behavior. These analyses highlight key words contributing to the model's predictions, providing a more interpretable view of how the model processes input text. While not perfect, these methods still offer useful insights into the model's decision-making process.

6. Deployment

The final model was exported to ONNX format, enabling efficient inference in production environments. This demonstrates that the model is not only effective but also deployable, with potential for real-time applications.

7. Limitations

Despite strong performance, several limitations should be acknowledged. First, training was limited to a small number of epochs due to computational constraints, which may prevent the models from reaching optimal performance.

Second, the preprocessing and augmentation techniques explored were relatively simple. More advanced approaches, such as back-translation or contextual augmentation, could yield different results.

Third, interpretability was primarily based on attention weight analysis. LIME was incorporated into the pipeline but could not be fully executed due to environment constraints. While informative, these methods do not fully explain model behavior.

Finally, the experiments were conducted on a single dataset, which limits the generalizability of the findings.

8. Future Work

Future work could explore more advanced data augmentation strategies, including paraphrasing using large language models. Additionally, experimenting with more complex classification heads and conducting more extensive hyperparameter tuning may further improve performance.

From an interpretability perspective, integrating SHAP would provide more comprehensive explanations. Finally, evaluating the models on multiple datasets would allow for a more robust assessment of cross-domain generalization.